PassFrame
tuition

Number Sequences

Year 5 Accelerated — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: What is the next number in the sequence: 4, 8, 16, 32, ?

Method & answer: Each number is doubled (×2). 32 × 2 = 64. Answer: 64.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. What is the next number in the sequence: 3, 4, 5, 6, ?
2. What is the next number in the sequence: 10, 20, 30, 40, ?
3. What is the next number in the sequence: 1, 4, 7, 10, ?
4. What is the next number in the sequence: 30, 27, 24, 21, ?
5. What is the next number in the sequence: 8, 16, 24, 32, ?
6. What is the next number in the sequence: 60, 54, 48, 42, ?
7. What is the next number in the sequence: 6, 12, 24, 48, ?
8. What is the next number in the sequence: 45, 40, 35, 30, ?
9. What is the next number in the sequence: 4, 12, 36, 108, ?
10. What is the next number in the sequence: 11, 22, 33, 44, ?
11. What is the next number in the sequence: 84, 72, 60, 48, ?
12. What is the next number in the sequence: 9, 18, 27, 36, ?
13. What is the next number in the sequence: 1, 5, 9, 13, ?
14. What is the next number in the sequence: 63, 56, 49, 42, ?
15. What is the next number in the sequence: 6, 11, 16, 21, ?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. What is the next number in the sequence: 25, 36, 49, 64, ?
2. What is the next number in the sequence: 8, 16, 32, 64, 128, ?
3. What is the next number in the sequence: 2, 8, 32, 128, ?
4. What is the next number in the sequence: 4096, 1024, 256, 64, ?
5. What is the next number in the sequence: 28, 36, 45, 55, ?
6. What is the next number in the sequence: 3, 10, 19, 30, 43, ?
7. What is the next number in the sequence: 9, 18, 36, 72, 144, ?
8. What is the next number in the sequence: 640, 320, 160, 80, 40, ?
9. What is the next number in the sequence: 4, 5, 7, 10, 14, 19, ?
10. What is the next number in the sequence: 5, 15, 45, 135, 405, ?
11. What is the next number in the sequence: 70, 59, 48, 37, ?
12. What is the next number in the sequence: 8, 13, 21, 34, 55, ?
13. What is the next number in the sequence: 81, 100, 121, 144, ?
14. What is the next number in the sequence: 7, 21, 63, 189, 567, ?
15. What is the next number in the sequence: 50, 43, 36, 29, ?

Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. What is the next number in the sequence: 6, 9, 7, 10, 8, ?
6. In this sequence, each number is formed by adding the digits of the
11. A sequence multiplies by 3 and then adds 1 each time: 2, 7, 22,

previous number to itself: 23, 28, 38, 49, ?. Check: 23 + (2+3) = 28, 28 + (2+8) = 38, 38 + (3+8) = 49. What is the next number?

67, ?. Check: 2×3+1=7, 7×3+1=22, 22×3+1=67. What comes next?

2. What is the next number in the sequence: 2, 4, 7, 11, ?

7. A sequence follows the rule 'double the previous number, then subtract 3': 4, 5, 7, 11, 19, ?. Check: 4×2−3=5, 5×2−3=7, 7×2−3=11, 11×2−3=19. What comes next?

12. Look at the pattern of gaps between these numbers: 4, 6, 10, 16, 24, ?. The gaps themselves increase by 2 each time (2, 4, 6, 8…). What is the next number?

3. What is the next number in the sequence: 3, 6, 5, 10, 9, ?

8. Two sequences are hidden in this list, one in the odd positions and one in the even positions: 5, 100, 8, 80, 11, 60, ?. The odd-position numbers increase by 3 each time; the even-position numbers decrease by 20 each time. What is the next number?

13. A sequence alternates between two rules: add 7, then multiply by 2: 3, 10, 20, 27, 54, ?. Check: 3+7=10, 10×2=20, 20+7=27, 27×2=54. What comes next?

4. What is the next number in the sequence: 100, 90, 81, 73, ?

9. A sequence is built like this: start with 2 numbers, then each new number is the sum of the previous two, minus 2: 3, 4, 5, 7, 10, ?. Check: 3+4−2=5, 4+5−2=7, 5+7−2=10. What comes next?

14. Each number in this sequence is the square of its position, plus 2: 3, 6, 11, 18, 27, ?. ($1^2+2=3$, $2^2+2=6$, $3^2+2=11$, $4^2+2=18$, $5^2+2=27$.) What is the 6th number in the sequence?

5. What is the next number in the sequence: 1, 2, 4, 8, 15, ?

10. The triangular numbers are formed by adding one more each time: 1, 3, 6, 10, 15, ?. (1, then 1+2, then 1+2+3, then 1+2+3+4, and so on.) What is the 6th triangular number?

15. A sequence works by subtracting an increasing odd number each time: 40, 39, 36, 31, 24, ?. Check: 40−1=39, 39−3=36, 36−5=31, 31−7=24. What comes next?